

Rectifying Varying Motion to Constant Motion

EID103

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Project Description

In nature and human designed mechanisms, there exist sources of motion that may be non-constant but unidirectional, such as waterfalls, rivers, wind and engines. Although this type of motion has utility, more predictable and constant motion can be more useful in more applications. Therefore, this project aimed to design a system to convert non-constant and unidirectional motion from an electrical motor into uniform, unidirectional motion.

More specifically, these are the given requirements and restraints:

1. The system must be assembled onto and off the aluminum breadboard in less than 5 minutes
2. The system must connect to the input motor shaft; this coupling device **MUST** have a gearing feature on it to mesh with the encoder gear.
3. The system must have a way to measure (either in real-time or through post processing) its input and output rotational speeds vs. time.
4. The output rotational motion **MUST** rotate a disc with a diameter between 1-3 inches.
5. Your system will be tested at 3 different frequencies 1Hz, 0.5Hz, 0.25Hz
6. The system must take less than 10 seconds to get to a steady state and no parts that spin in excess.

Design Process

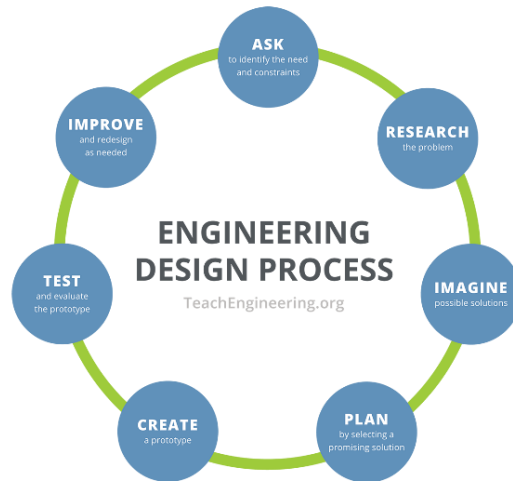


Figure 1. teachengineering.org [1]

Three possible designs of the motion rectifier were initially proposed, namely *Jeremy*, *Chris*, and *Antoine*.

Prototype 1

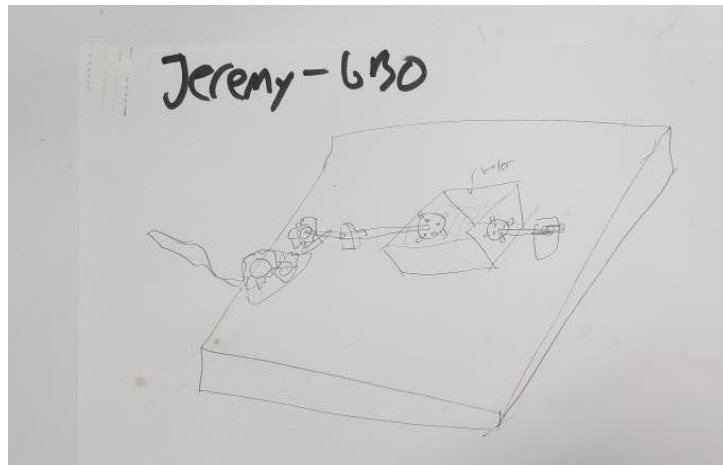


Figure 2. *Jeremy*

Jeremy uses water and magnets to reduce the velocity of the rod, without changing the inertia of the system. The main component of the design is inside of a water tank, where magnetically coupled wheels decouple at high velocity, setting a cap on the highest velocity

achieved. Fins placed along the circumference of the wheel further slow the rotation. The design rectifies the irregular motion by limiting the angular velocity of the system.

Prototype 2

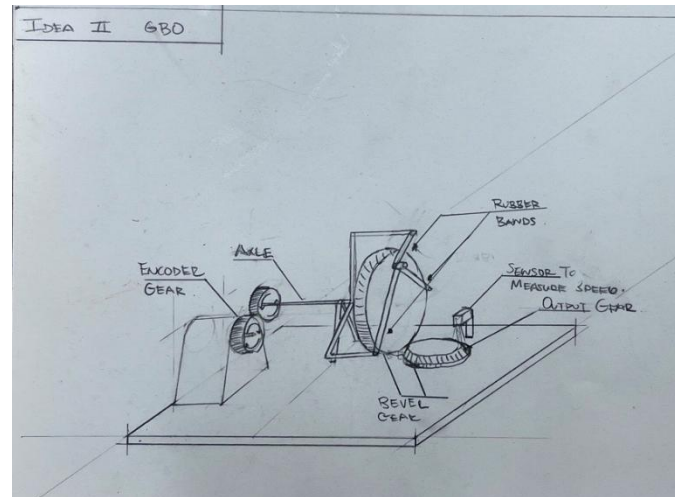


Figure 2. Chris

Chris is an elastic resistance-based rectifier, using rubber bands to prevent the rotation of an off-center peg on a wheel. When the rod is rotated it rotates the wheel as well. The rubber bands are attached to the off centered peg, this way any rotation in of the wheel is resisted by the rubber bands held by separate arms, rectifying the non-constant angular velocity.

Prototype 3

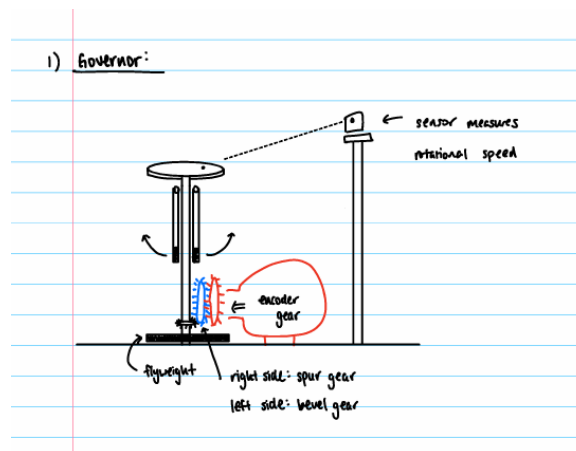


Figure 3. Antoine

Antoine is an inertia-based motion rectifier whose inertia increases as the angular velocity increases. The governor spreads out as the rod it is attached to increases in velocity, along with marbles inside of a flywheel moving towards the radius. The two mechanisms work in tandem to increase the inertia of the system based on speed, thus making it an effective solution for motion rectifying.

Table 1 displays the decision matrix that was used to determine that *Antoine* was to be the final design. Onshape was used to CAD the parts of the design, and a Prusa Mini was used to 3D print all parts. The initial base and governor were too weak to effectively rectify the velocity of the motor. Through Onshape, the team modeled a base and a flywheel containing ball bearings, printed by the Prusa Mini. The testing revealed that the rod does not spin fully axially, meaning that it sways. The team once again redesigned the base with widened footprints, stronger and higher columns, and the flywheel with more cavities to house a greater mass of marbles. The cycle delivered a substantially heavier base and flywheel, and the added inertia finally capped rod speed at an acceptable level.

Designs	Difficulty	Cost	Durability	Ease of Dissassembly	Predicted Effectiveness	Point Totals
Jeremy	2	5	3	1	4	15
Chris	8	8	2	8	5	31
Antoine	5	7	9	6	9	36

Table 1. Decision Matrix

Design Analysis

Centrifugal Forces

Antoine used the fundamental principle of centrifugal forces; in essence, as the machine spins faster, the tendency of components to fly out of the path of rotation increases. The flyball governor harnesses this effect through two arms with weights at its ends. As rotational speeds increase, the arms create circular paths with increasing radius. Weight concentrated at greater radii increases the systems' moment of inertia, effectively slowing down the angular speed. Similarly, weighted ball bearings were placed in channels and pockets of the flywheel. As rotational speeds increase, the ball bearings experience outward radial forces, enabling them to travel in greater radii. The moment of inertia is again increased, reducing the angular speed of the system.

The Governor

The governor integrates more than just two arms. Two shorter, control arms are required to modulate the speed that the longer, weighted arms ascend at – having the weighted arms fly vertically at unregulated speeds is undesirable. To implement the control arms, a collar was incorporated to ensure fluid, but stable motion. The collar secured the control arms and was constrained to a vertical path, only allowing upward and downward sliding along the shaft. After tests, the team added hooks on the collar and arms to accommodate rubber bands. Initially, the rate at which the arms ascended was too high and locked out at intermediate speeds. To solve this, rubber bands introduce further resistance, requiring higher speeds to raise the weighted arms to the same height.

The last part of the governor is the governor top. The top anchors the weighted arms in place and provides a datum, or a reference point, to take measurements from. A black dot

marked on the governor top allows the angular velocity of the system to be tracked through Tracker. The governor top is fastened to the top of the shaft through a collar.



Figure 4. Flyball Governor Mounted on Shaft

Power Transfer

The motor is mounted with a horizontal axis of rotation, but the shaft is mounted with its axis vertically. To transfer all rotational motion of the motor to the shaft, the team decided to use spur and bevel gears. Two 45° bevel gears converted horizontal rotation to a vertical analog; however, the bevel gears did not mesh with the encoder gear, so a combined spur-bevel gear was modeled. The spur gear delivers the power, and the bevel gear directs that power 90° . The combined spur-bevel gear was mounted on the motor and the singular bevel gear was fastened onto the shaft using a collar.



Figure 5. Collared Bevel Gear



Figure 6. Combined Spur-Bevel Gear

The Base

Given an 8 mm-diameter metal shaft, the team designed a base that connects two platforms through four columns. The shaft went through both platforms with a bearing on each, which were located at different heights. The platforms were fixed together through the columns, with bolts and nuts fastening them together. Two platforms were secured at different heights to ensure stability, and the bottom platform was secured to the breadboard using bolts. A prominent issue was that the breadboard and shaft were both made of metal and in direct contact, producing substantial friction. To prevent this, a third platform was modeled and assembled under the main base structure so that the contact was metal-on-plastic – not an ideal solution, but one that eliminates a large amount of the friction present.



Figure 7. Base

Simple Machines

From an engineering perspective, this design exploited the classic wheel-and-axle. The flywheel was fastened to the metal shaft and the combined gear was operated through its own shared axle. Both subsystems served different purposes, despite the simple machine commonality. The flywheel used the large wheel to store rotational kinetic energy, ultimately aiming to attenuate the oscillating input speed. The gear system used the “wheel”, or the gear, to reorient the axis of rotation.

System Diagram

Red arrows indicate rotational power transfers:

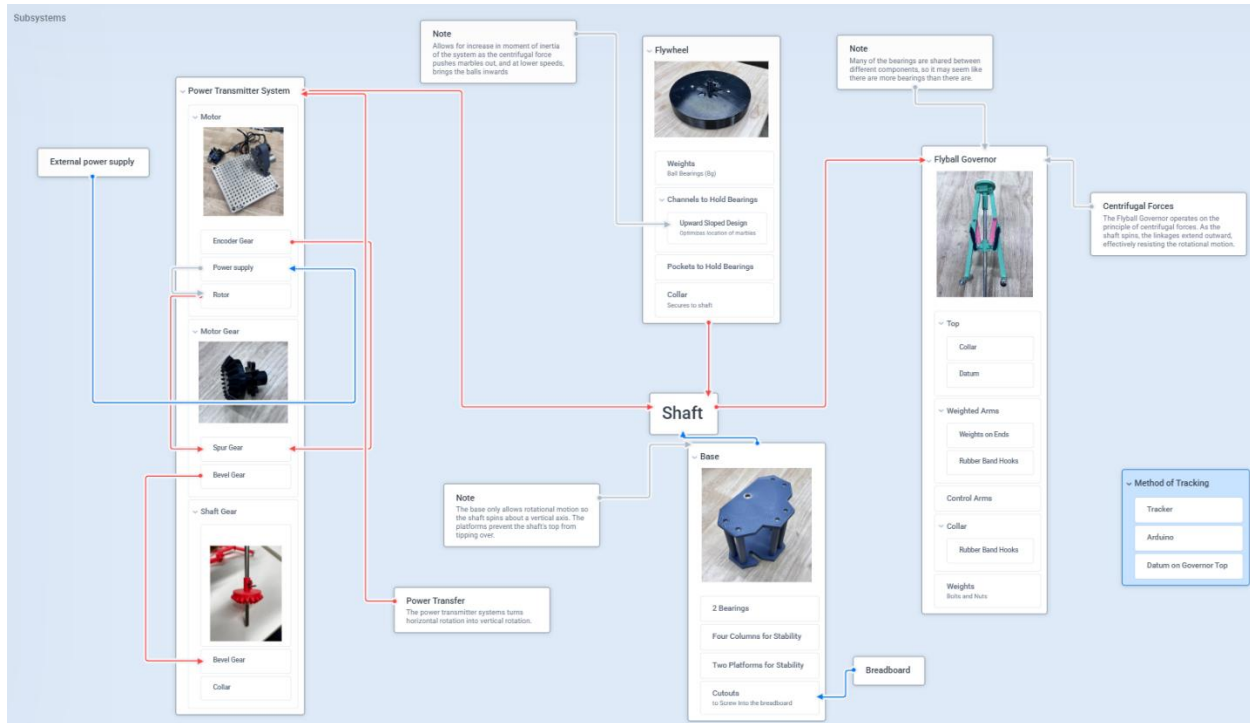


Figure 8. System Diagram

Bill of Materials

Subsystem	Component	Quantity	Cost	Brief Description
Power Transfer	Spur-Bevel Gear	1	--	Combined gear that redirects rotational moment 90 degrees.
	Bevel Gear	1	--	Gear that the shaft relies upon for rotation.
Flywheel	Flywheel	1	--	Holds the ball bearings and operates on centrifugal forces.
	Flywheel Cover	1	--	Placed on flywheel to prevent weights from flying out.
	Ball Bearings	60	\$8.99	Weights that increase the inertia of the system.
	Screws	3	--	Secures the flywheel cover to flywheel
	Nuts	3	--	Secures the flywheel cover to flywheel
Base	Collar	1	--	Secures the flywheel to the shaft.
	Platforms	3	--	Provides stability when the shaft rotates.
	Columns	4	--	Provides stability when the shaft rotates.
	Bolts	10	--	Secures the platforms to the breadboard; secures the columns to the platforms.
	Nuts	8	--	Secures the columns to the platforms.
Flyball Governor	Bearings	2	--	Allows the shaft to freely rotate through the platforms.
	Governor Top	1	--	Holds the weighted arms in place; provides a datum for measurements.
	Weighted Arms	2	--	Hold weights on its ends; operates on centrifugal forces.
	Control Arms	2	--	Controls the arms' rate of extension and retraction from the center of rotation.
	Rubber Bands	2	--	Provides resistance to the arms' rate of extension from the center of rotation.
	Screws (Weight)	2	--	Provides the weight for the arms.
	Nuts (Weight)	16	--	Provides the weight for the arms.
	Screws	7	--	Secures the arms to each other, governor top, and bushings, and collar.
	Nuts	7	--	Secures the arms to each other, governor top, and bushings, and collar.
	Bushings	6	--	Reduces friction created from the moving arms.
Shaft	Collar	2	--	Collar holds the control arms at a fixed location to prevent variance.
	Metal Shaft	1	--	Unifies all subsystems together; source of rotation for the flywheel and governor.

Table 2. Bill of Materials

Iterative Design

Sliding Collar

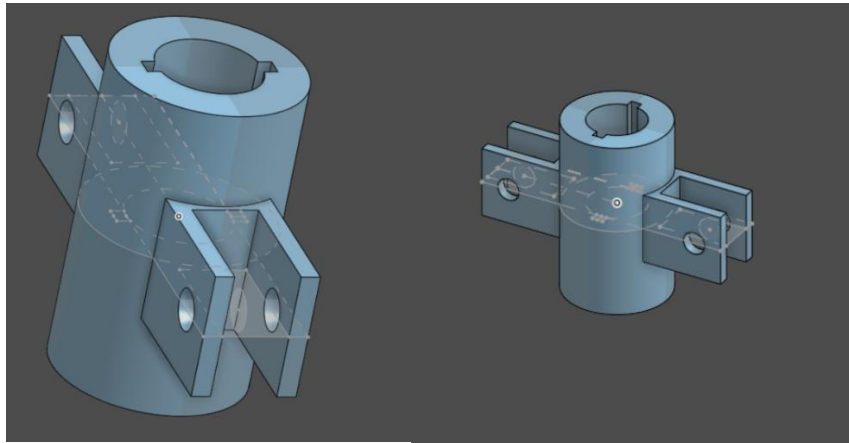


Figure 9a. Collar V1

This was the initial design of the collar, very basic in its function. There are two flanges with holes so that a M3 screw could slide through and hold one end of the short arm. Note the thinness of the flanges as when printing, the team noticed that the thickness was only about two passes of filament.

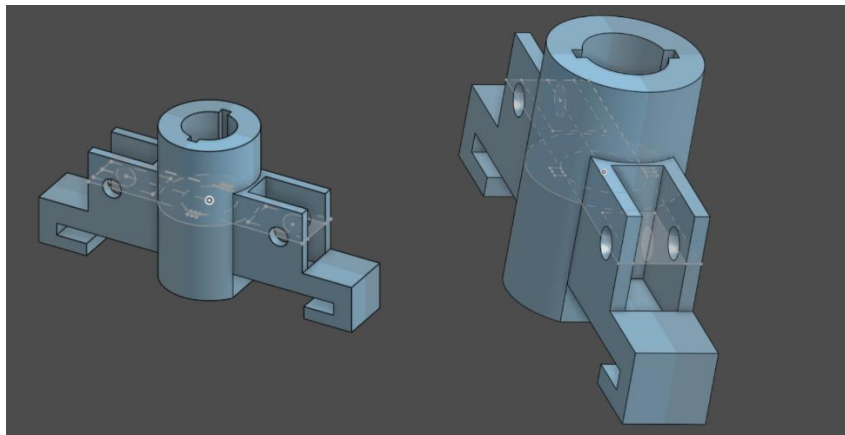


Figure 9b. Collar V2

This was version two, where the team added clips to the end to be able to hold the rubber band mechanism since the system was still not attenuating enough. The flange thickness was not changed but was still noted to change in the final version.

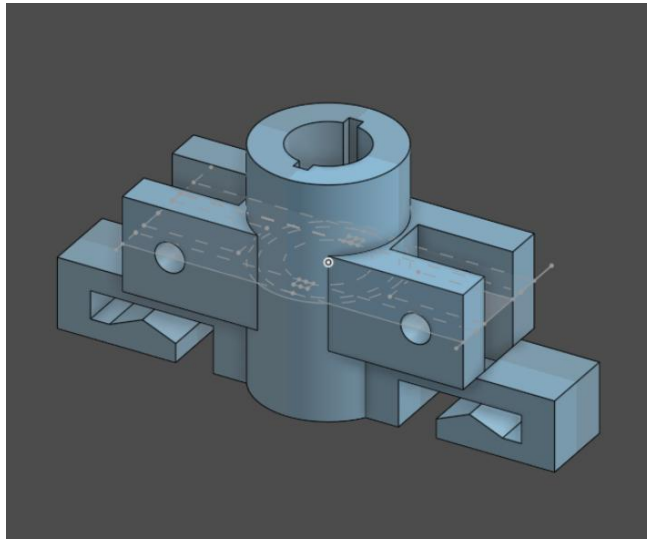


Figure 9c. Collar V3

This was the final version of the collar. Notice the added thickness to a final 4mm for the flanges while also being attached to the clip for extra support. The length of the clips was also increased to extend outwards more after testing version 2, since the arms would lock up due to the pulling of the rubber bands. This enables the arms to be held under a greater proportion of vertical load rather than horizontal to prevent locking up.

Long Arm



Figure10a. Long Arm V1

This was the original design for the longer arm; note that apart from the three connecting holes, there was a lack of features. The farthest hole on the right is 1/4" diameter to accommodate the 3" long bolt with nuts acting as a weight.

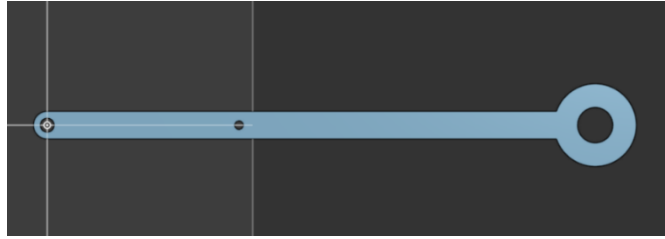


Figure 10b. Long Arm V2

Version 2 only saw an increase in the hole size at the end as the team wanted to solve the lack of attenuation by increasing the weight of the bolt at the end, changing it to a ½” diameter. Consequently, it caused the arm to lose much more stability and eventually resulted in a change in version 4.

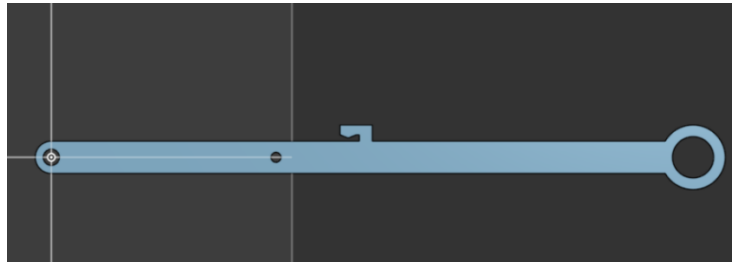


Figure 10c. Long Arm V3

Version 3 saw the addition of the clip to hold the rubber band, like the feature added on the collar. Note the thinning out of the hole at the right side to decrease bulk, but ultimately was unstable.

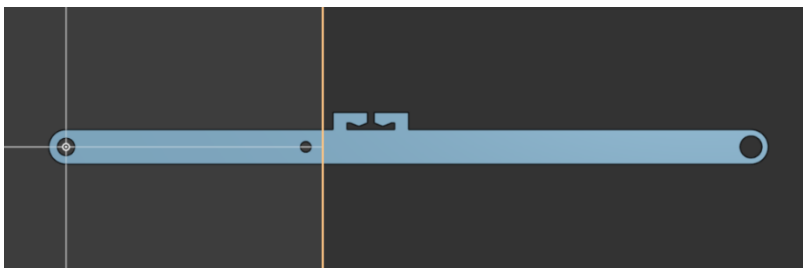


Figure 10d. Long Arm V4

The final version saw the return of the 1/4” diameter bolt and an addition of another clip to the left side. This was due to the rubber band slipping along the arm as the arm moved up during rotation.

Governor Top

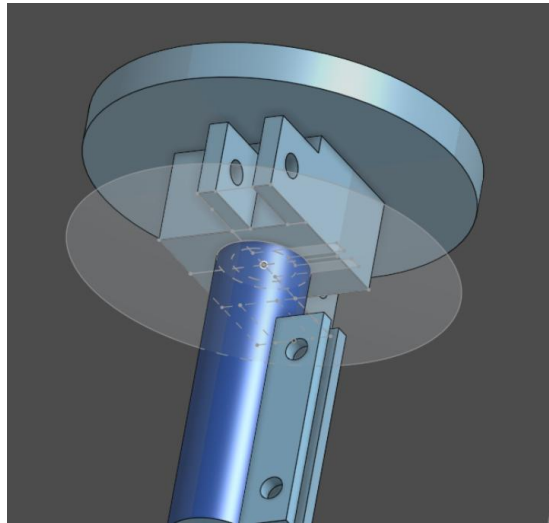


Figure 11a. Governor Top V1

The governor only received one set of changes during the entire process. Note the thickness of the flanges and the length of the sleeve below it. The thickness of the flanges were too thin, like those on the collar and the length of the sleeve was too short, causing the arms to lock up in the highest position.

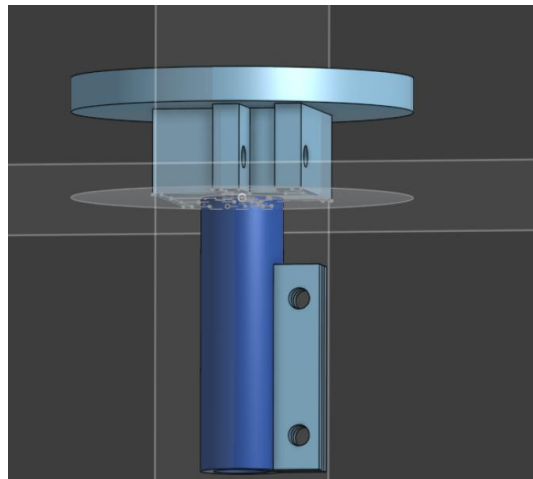


Figure 11b. Governor Top V2

This was the final version of the governor, note the extension of the flanges up to the top for extra stability and an increase in thickness as well. The sleeve length was also increased so that the system would not lock up by preventing the collar from reaching a higher position.

Base

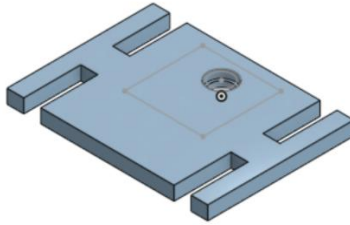


Figure 12a. Base V1

This was the original design for the base, with slots large enough to compensate for any miscalculation in the distance between holes, and a hole for the bearing. The design was created before knowing that two bearings were required, thus the presence of only one hole.

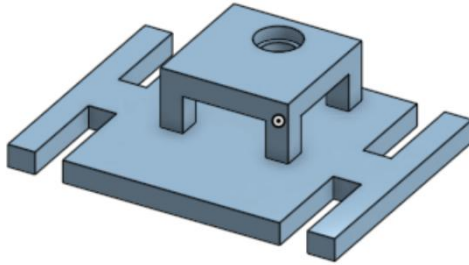


Figure 12b. Base V2.

The second iteration attempted to solve the problem of only having one bearing. However, due to the low difference in height between the two bearings, the mechanism went out of alignment and spun uncontrollably.

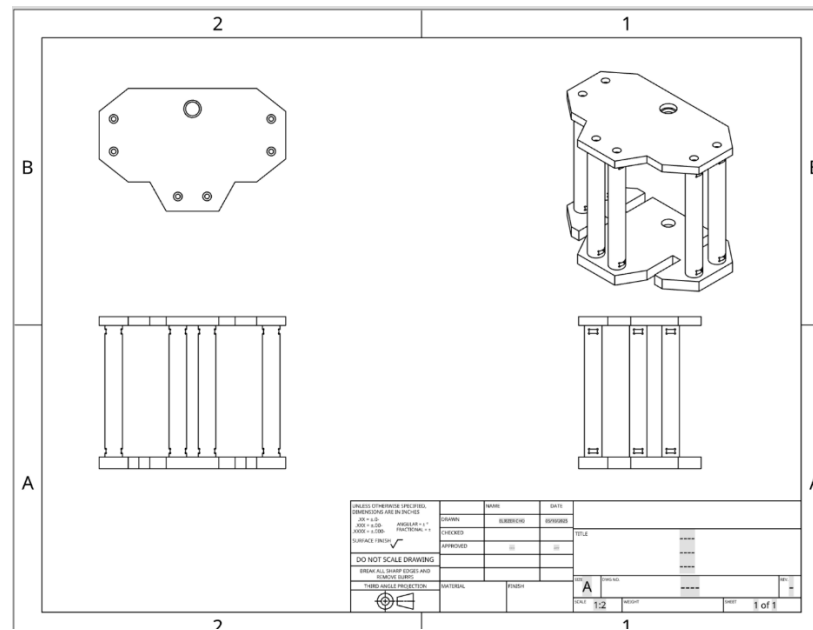


Figure 12c. Base V3

The third iteration was also the final version, where a second layer was added. By adding the second layer, the stability of the whole system improved; the team incorporated a second bearing at the top to also prevent rubbing of parts. Secondly, when installing the base onto the breadboard the team needed to add an old base plate under the hole where the rod would insert to prevent the rod from scratching the breadboard and causing more friction.

Flywheel

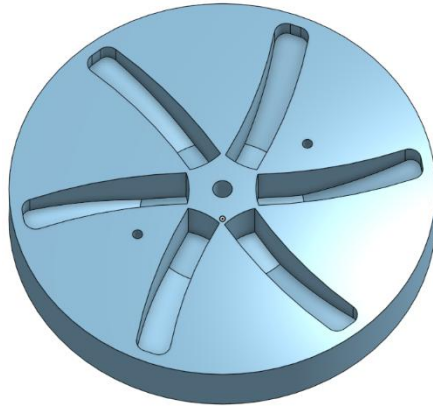


Figure 13a. Flywheel V1

When the first flywheel was made, six channels were created for the ball bearings to sit in. The channels were modeled so that they sloped upwards, so a ball bearing at a greater radius sat at greater height. While this wasn't exactly calibrated, the idea is the upward slope resisted the centrifugal forces experienced ball bearings. This allowed the ball bearings to travel outwards at controlled speeds.

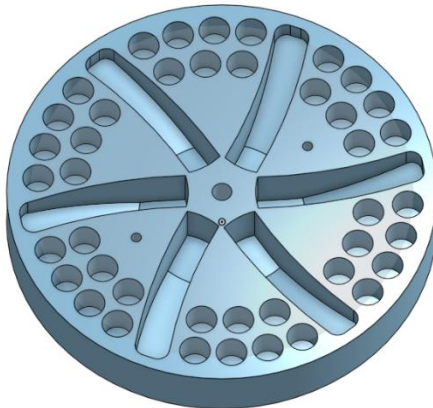


Figure 13b. Flywheel V2

After testing, inertia was determined to be the most influential factor for attenuating oscillations. Therefore, more mass was added to the system. The optimal method was to increase the weight of the flywheel by creating pockets that ball bearings could sit in. These pockets were placed as close to the edge of the flywheel as possible to maximize the rotational inertia of the subsystem, aimed to decrease the angular velocity.

Exploded View of Final Assembly

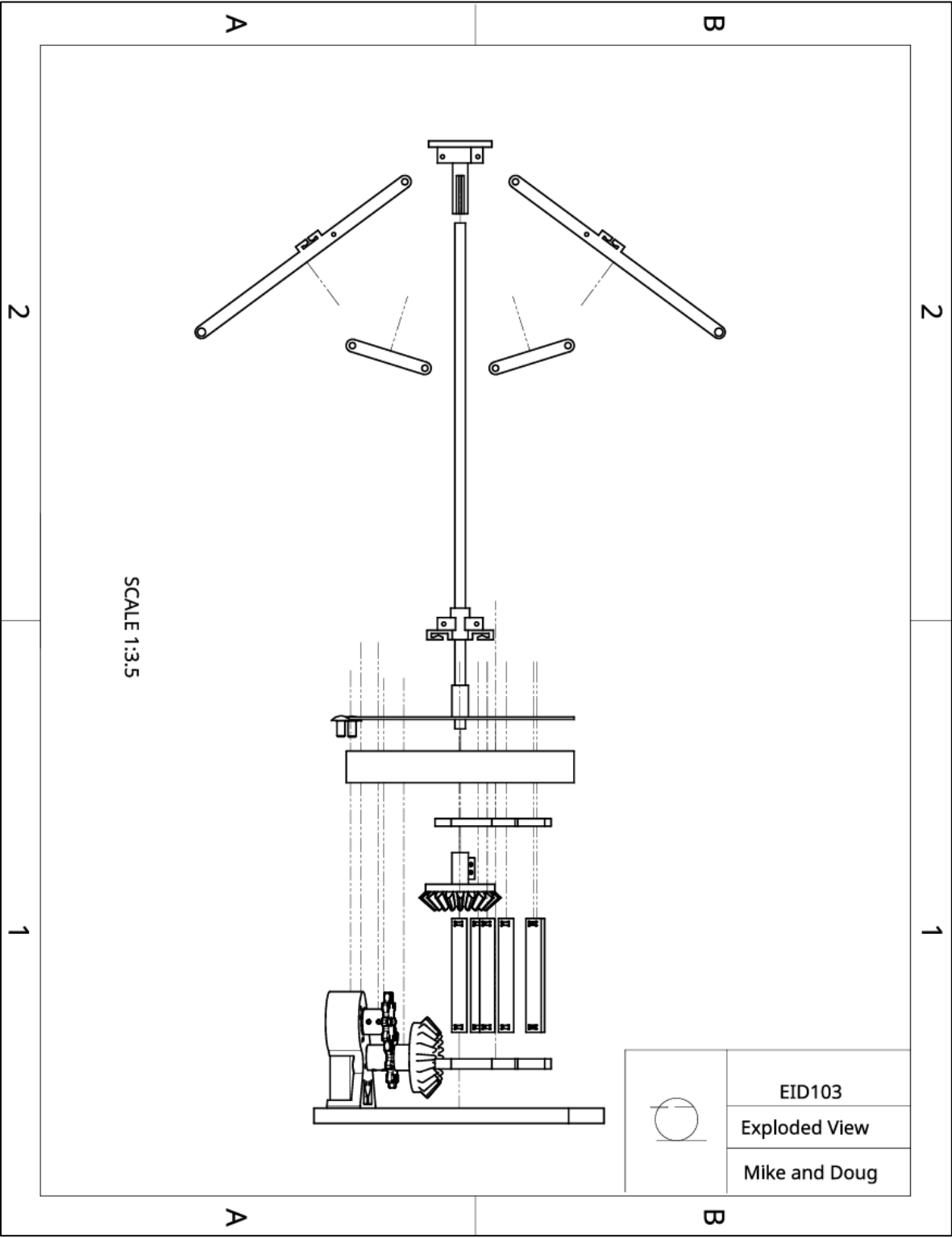


Figure 14. Exploded View of Assembly

Orthographic Projection of Final Assembly

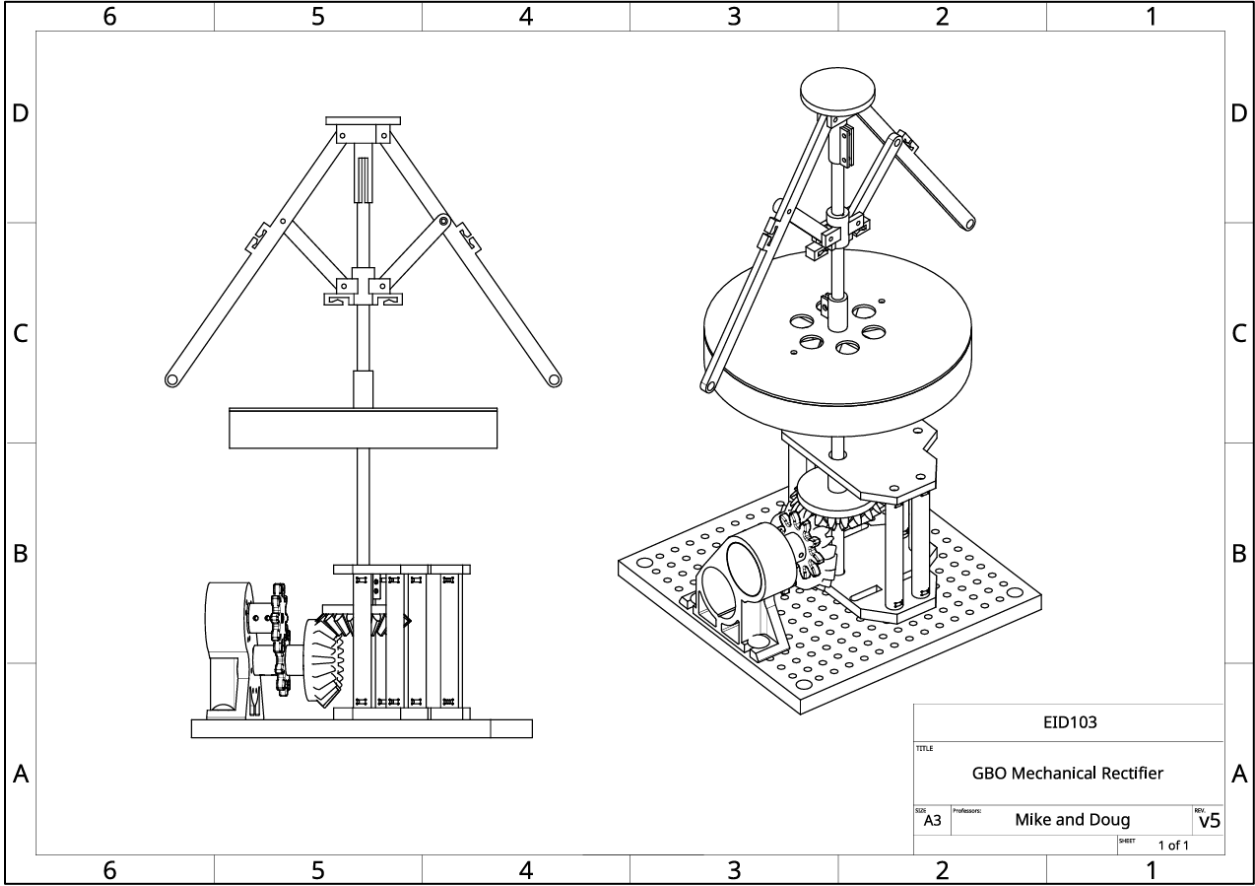


Figure 15. Orthographic Projection of Assembly

Outcome Analysis

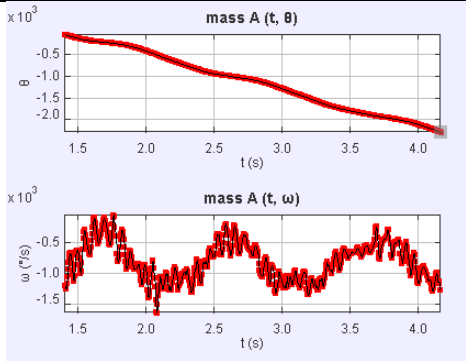
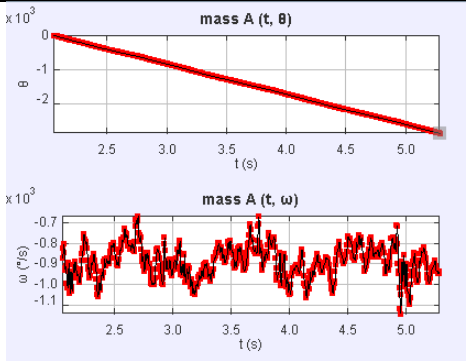
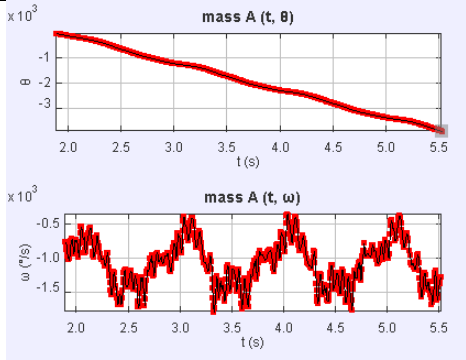
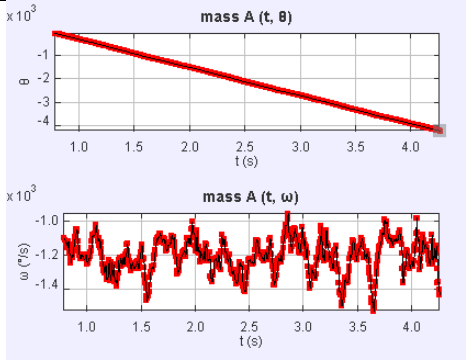
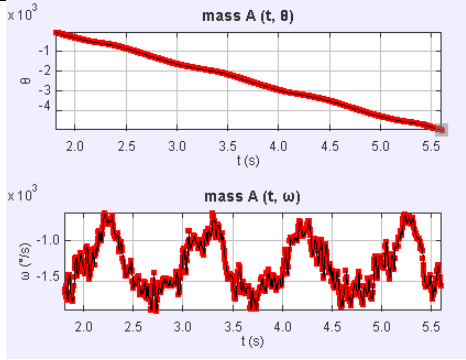
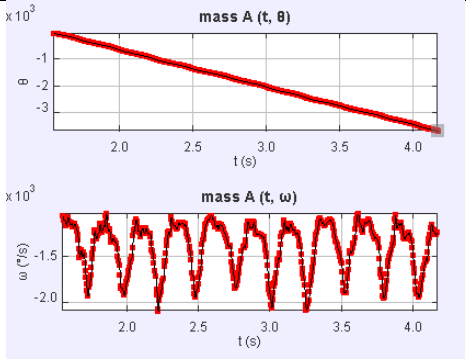
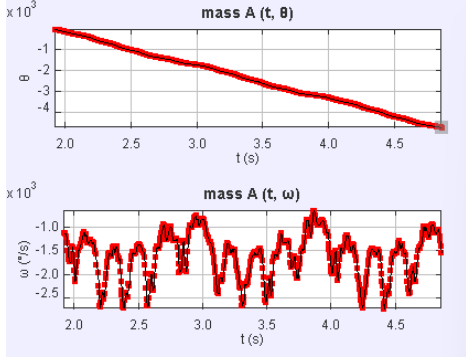
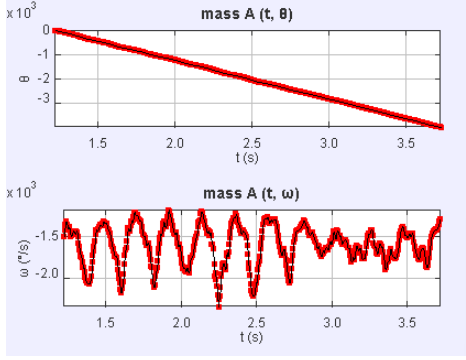
Voltage	Not Rectified	Rectified
6V		
7V		
8V		
9V		

Table 3. Unrectified and Rectified Graphs at Four Voltages, Tracker

Table 3 above displays data collected through Tracker. Theta vs. time graphs and angular velocity vs. time graphs at four different voltages are shown. The graphs on the left provide data of the unrectified system and the graphs on the right provide data on the rectified system.

Just by looking at the theta vs. time graphs, the governor and flywheel have an evident effect on dampening the oscillating rotational velocity of the motor. All the unrectified graphs display nonconstant, varying slopes whereas all the rectified graphs present near-constant linear curves. This is within reason – the subsystems effectively made the rate of change of the angle constant, smoothening out the line of the theta vs. time graphs.

The angular velocity vs time graphs exhibit a similar trend. Especially for plots 6V through 9V, the unrectified data mimics oscillatory graphs – the curve of best fit is a sinusoid. By contrast, the rectified graphs, more obviously seen at 6V and 7V, the line of best fit can be assumed to be adequately horizontal. This implies that the oscillations were attenuated.

It is important to note that Tracker plotted out curves that are not exactly horizontal nor smooth because the videos being tracked were taken from a bird's-eye view. From this perspective, the vibrating nature of the shaft is particularly apparent. Furthermore, videos were taken by hand, which were unsurprisingly shaky. Since the video was not centered frame by frame, the axes and points that the auto-tracking system followed are flawed, producing erroneous results. However, the approximate shapes of the graphs can be inferred and analyzed to come to general conclusions.

Using these graphs and the decibel scale, the attenuation of the subsystems can be quantified. For the most favorable outcome, the 6V graphs will be considered. Its percent change in amplitude of the angular velocity vs. time graphs from the unrectified to rectified plots are

greatest. The amplitude of the unrectified graph is about 0.8 and the *approximate average* amplitude of the rectified graph is about 0.1. Note that these values are scaled down by a factor of 10E3. The sample calculation is as follows:

$$dB = 20 * \log_{10} \left(\frac{A_1}{A_0} \right) = 20 * \log_{10} \left(\frac{0.1}{0.8} \right) \approx -18.06$$

If the system were to not have affected the angular velocity, the ratio of amplitudes would yield 1, producing a calculation of 0 dB attenuation. An ideal value obtained should be negative infinity, as the desired rectified amplitude is 0. The following are approximate values of the remaining voltages, and again, the amplitudes are scaled down by a factor of 10E3:

Voltage	Amplitude (NOT Rec.)	Amplitude (Rec.)	Decibel Scale
7V	0.65	0.2	-10.24
8V	0.75	0.5	-3.52
9V	0.90	0.35	-8.20

Table 4. Decibel Scale Calculations for Remaining Voltages, Tracker

The team also measured the angular velocity of the system using Arduino (Table __). A microprocessor measured the quantities and output them onto the serial monitor. The same calculations were performed to determine the decibel scale. To derive the amplitude, differences between relative minima (yellow) and relative maxima (orange) within a time interval were halved.

Voltage	Amplitude (NOT Rec.)	Amplitude (Rec.)	Decibel Scale
6V	2.90	0.68	-12.60
7V	3.07	0.63	-13.76
8V	3.42	0.59	-15.26
9V	3.365	0.57	-15.4

Table 5. Decibel Scale Calculations, Arduino

Rectified 6V	Unrectified 6V	Rectified 7V	Unrectified 7V	Rectified 8V	Unrectified 8V	Rectified 9V	Unrectified 9V
15.83	15.59	20.54	20.04	24.16	23.78	28.17	23.9
15.83	15.54	20.51	20.03	24.16	23.69	28.12	23.81
15.82	15.5	20.53	19.98	24.11	23.61	28.08	23.77
15.87	15.56	20.5	20.07	24.16	23.69	28.04	23.84
15.91	15.62	20.65	20.12	24.15	23.65	28.05	23.8
15.85	15.88	20.66	20.16	24.25	23.68	28.06	24.03
15.89	15.92	20.71	20.3	24.24	23.7	28.07	24.14
15.94	16.01	20.72	20.33	24.29	23.77	28.08	24.19
15.93	16.1	20.78	20.51	24.36	23.88	28.14	24.39
15.97	16.33	20.74	20.69	24.39	24.59	28.2	24.53
15.95	16.35	20.8	20.8	24.43	24.72	28.2	24.71
16.04	16.53	20.91	20.96	24.46	24.89	28.31	24.83
16.22	16.85	21.06	21.58	24.5	25.1	28.31	25.1
16.2	17.05	21.11	21.81	24.48	25.26	28.36	25.21
16.23	17.24	21.25	21.97	24.56	25.51	28.48	25.51
16.25	17.37	21.24	22.07	24.54	25.69	28.47	25.59
16.33	17.59	21.27	22.37	24.57	25.92	28.51	25.88
16.3	17.86	21.36	22.46	24.59	26.09	28.55	26.1
16.37	18.01	21.34	22.69	24.62	26.4	28.59	26.3
16.49	18.15	21.47	22.86	24.64	26.54	28.57	26.55
16.64	18.44	21.5	22.97	24.67	27.12	28.71	26.79
16.6	18.56	21.47	23.18	24.78	27.43	28.69	26.86
16.65	19.52	21.6	24.75	24.8	27.58	28.72	27.14
16.71	19.74	21.67	24.66	24.81	27.76	28.75	27.79
16.76	19.85	21.69	23.69	24.88	28.04	28.73	27.86
16.76	20	21.76	23.91	24.89	28.31	28.81	28.46
16.81	20.15	21.72	24.07	25.01	28.46	29.05	28.6
16.96	20.24	21.73	24.18	25.01	28.61	29.06	28.79
17.07	20.33	21.74	24.38	25.07	28.79	29.12	29.25
17.06	20.51	21.7	24.52	25.17	28.92	29.08	29.31
17.09	20.53	21.72	24.7	25.17	29.04	29.15	29.46
17.13	20.65	21.83	25.1	25.22	29.21	29.15	29.61
17.11	20.67	21.79	25.15	25.21	29.32	29.21	29.75
17.14	20.73	21.75	25.3	25.26	29.66	29.18	29.93
17.18	20.9	21.76	25.39	25.25	29.7	29.14	30
17.16	20.95	21.77	25.43	25.25	29.88	29.14	30.07
17.14	21	21.83	25.67	25.24	29.91	29.15	30.19
17.12	21.05	21.79	25.7	25.29	30.03	29.11	30.3
17.15	21.09	21.75	25.67	25.28	30.15	29.17	30.45
17.13	21.25	21.76	26.09	25.27	30.21	29.18	30.5
17.11	21.25	21.67	26	25.22	30.28	29.13	30.44
17.04	21.29	21.58	26.11	25.21	30.44	29.14	30.39

Table 6. Angular Velocity Data, Arduino

The decibel scale derived from the Arduino provides trends contrary to that of Tracker. This may be due to the vibrations making the Tracker values less robust; in contrast, this does not create as much noise on the encoder gear. At higher voltages, vibrations are more apparent, resulting in deceiving amplitudes and making the decibel scale values at 8V and 9V unreliable. Through the Arduino data, the system running at 9V attenuates the oscillations best.

Conclusion and Future Work

For future improvement, the team could decrease the danger to those who get too close, by further increasing the mass of the flywheel the inertia of the system can be increased even before the governor expands. The newly increased mass not only makes it safer for the observer because the speed at which the governor will be rotating will decrease, it also helps in further rectifying the motion of the motor and further allows for stability.

This increase in weight would need to be superseded by other various improvements. First, the thickness of flanges, arms and the size of screws used would have to increase. For example, increasing the bolts to M5 or M8 bolts for greater strength. Secondly, the tolerances of the features and mechanisms would have to become tighter which would increase the stability of the system during rotation. This could be implemented by using CNC mills to machine parts out of aluminum or use resin 3D printers to print more accurately. Third, the addition of more open caged bearings or oiled washers between surfaces to allow for less friction between parts. This would allow for more effective attenuation as less energy would be lost to friction, and the actual subsystems/mechanisms would not be prevented from adjusting for the varying motion. The team would have to change the CAD for the subsystems, adding channels in the flanges and on the sides of the arms to accommodate for bearings. Additionally, instead of using bolts, a bronze rod with collars to prevent sliding horizontally would decrease the friction even more.

Alternatively, the team could improve by adding a different powertrain system to decrease the overall speed of the gears; other teams incorporated designs such as using rubber bands to transfer the rotation from the driving gear to the rest of the system.

Though, overall, the team did well consider constraints on time, money and tools available to create a system that attenuated non-uniform motion. More specifically, by analyzing the Arduino data, at 9V, the rectified angular velocity range was 1.14 deg/s where the unattenuated range was 6.73 deg/s. This shows a significant effect on the angular velocity, making the motion much more consistent and uniform. Even the least effective experimental values for attenuation resulted in the range of angular velocity being 1.36 deg/s when the unattenuated values had a range of 5.79 deg/s.

Consequently, through the process of this project, the team learned about ideas such as mechanical advantage, considering strength of materials, manufacturing defects, drawbacks of certain methods of manufacturing, friction within systems and iterating on ineffective design choices.

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